(iv) **To solve an equation of the form** $a\cos\theta + b\sin\theta = c$:

$$\begin{aligned}
\text{Take } a &= r\cos\alpha,\ \ b = r\sin\alpha.\ \text{Then } r = \sqrt{a^2+b^2};\ \ \tan\alpha = \frac{b}{a},\ a \neq 0\\
a\cos\theta + b\sin\theta &= c \Rightarrow r\cos\alpha\cos\theta + r\sin\alpha\sin\theta = c\\
r\cos(\theta-\alpha) &= c\\
\cos(\theta-\alpha) &= \frac{c}{r} = \frac{c}{\sqrt{a^2+b^2}} = \cos\phi \ \text{(say)}\\
\theta-\alpha &= 2n\pi \pm \phi, n \in \mathbb{Z}\\
\theta &= 2n\pi + \alpha \pm \phi, n \in \mathbb{Z}.
\end{aligned}$$

The above method can be used only when $c \leq \sqrt{a^2+b^2}$.
If $c > \sqrt{a^2+b^2}$, then the equation $a\cos\theta + b\sin\theta = c$ has no solution.

Now we summarise the general solution of trigonometric equations.

Trigonometric Equation	General solution
$\sin\theta = 0$	$\theta = n\pi; n \in \mathbb{Z}$
$\cos\theta = 0$	$\theta = (2n+1)\frac{\pi}{2}; n \in \mathbb{Z}$
$\tan\theta = 0$	$\theta = n\pi; n \in \mathbb{Z}$
$\sin\theta = \sin\alpha$, where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$\theta = n\pi + (-1)^n\alpha, n \in \mathbb{Z}$
$\cos\theta = \cos\alpha$, where $\alpha \in [0, \pi]$	$\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$
$\tan\theta = \tan\alpha$, where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$\theta = n\pi + \alpha, n \in \mathbb{Z}$

Example 3.42 Find the principal solution of (i) $\sin\theta = \frac{1}{2}$ (ii) $\sin\theta = -\frac{\sqrt{3}}{2}$
(iii) cosec$\theta = -2$ (iv) $\cos\theta = \frac{1}{2}$

Solution:

(i) $\sin\theta = \frac{1}{2} > 0$ so principal value lies in the I quadrant.
$\sin\theta = \frac{1}{2} = \sin\frac{\pi}{6}$
Thus, $\theta = \frac{\pi}{6}$ is the principal solution.

(ii) $\sin\theta = -\frac{\sqrt{3}}{2}$